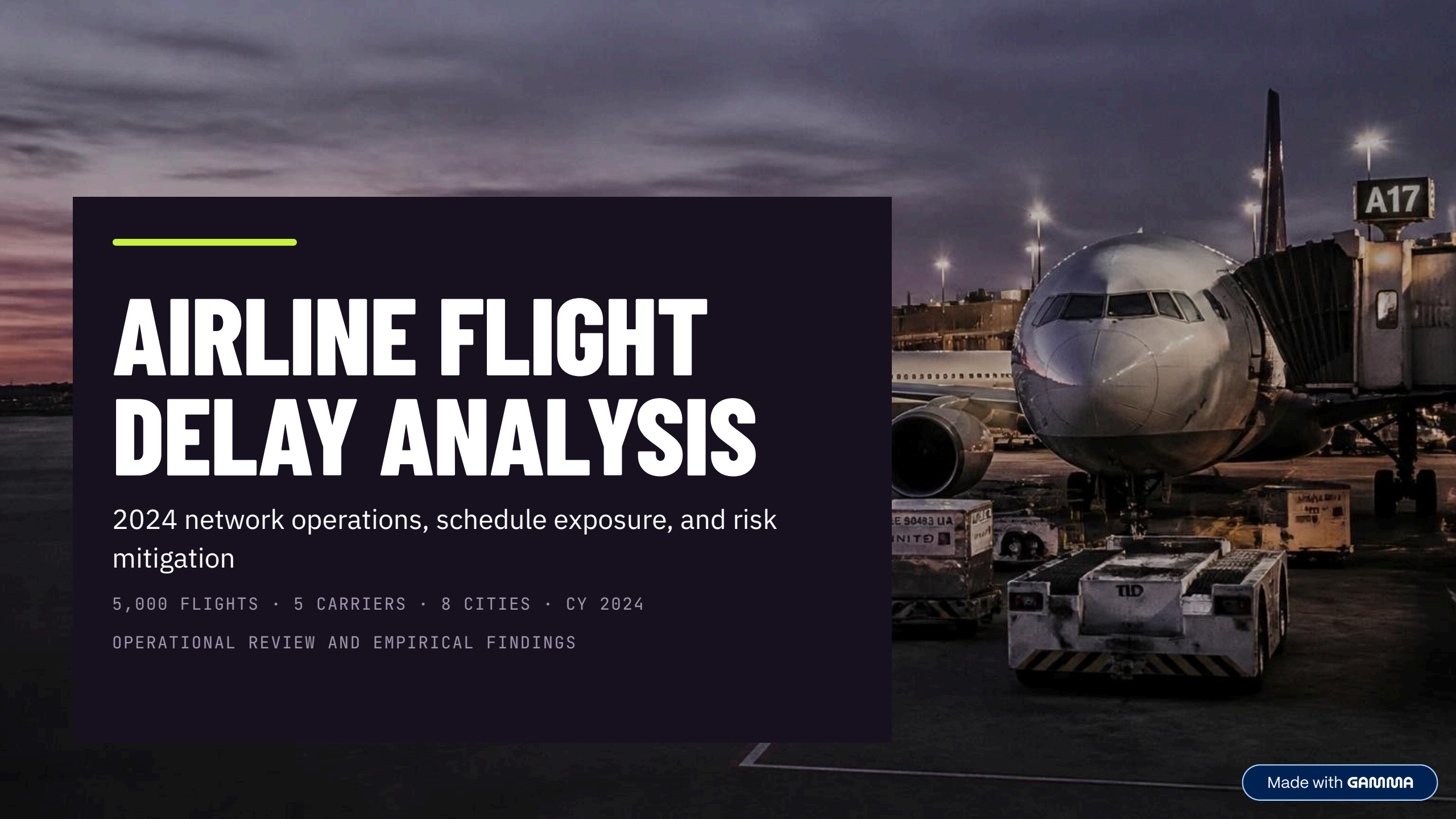


AIRLINE FLIGHT DELAY ANALYSIS

2024 network operations, schedule exposure, and risk mitigation

5,000 FLIGHTS · 5 CARRIERS · 8 CITIES · CY 2024

OPERATIONAL REVIEW AND EMPIRICAL FINDINGS

Made with GAMMA

EXECUTIVE SUMMARY

2024 OPERATIONAL BASELINE ACROSS FIVE CARRIERS

The annual dataset captures 5,000 flights operating across eight Indian commercial centers, with ₹4.31 billion in reported revenue. Operational performance varied by airline and operating hour, while load factors and passenger ratings were relatively similar.

SAMPLE VOLUME	AVG ARRIVAL DELAY	AVG LOAD FACTOR	TOTAL REVENUE
5,000	16.84m	82%	₹4.31B
● 8 CITIES · 5 AIRLINES	▲ NETWORK AVERAGE	● 81–82% BAND	REPORTED TOTAL

Environmental & temporal concentration

Weather and evening departures stand out in the analysis; observed delays were higher in storms and later in the day.

Commercial decoupling

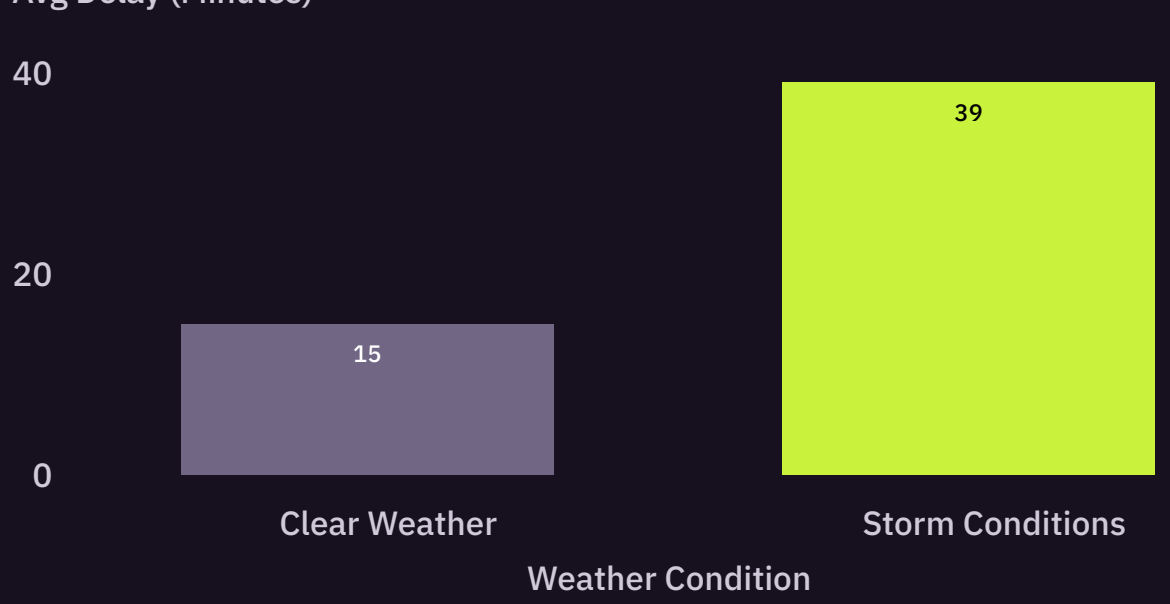
Customer satisfaction scores averaged ~3.7 out of 5 across all fleets, showing virtually no correlation (r = 0.01) with measured flight delays.

Made with GAMMA

ENVIRONMENTAL FACTORS

WEATHER AND TIMING SHAPE DELAY EXPOSURE

The source describes delays building through the operating day, with peaks in the evening and higher averages in storm conditions.



Storm delays averaged 2.6× those in clear weather (39 min vs. 15 min). Consider scheduling buffers and disruption readiness.

HOURLY DELAY PEAKS

25 MIN

▲ PEAK AT 9:00 PM

A secondary peak reached 24 minutes at 7:00 PM.

Analytical caveat

These figures reflect observed historical associations, not proof of causation. Scheduling and disruption measures should be validated.

Made with GAMMA

CARRIER BENCHMARK

CARRIER PERFORMANCE PROFILES ACROSS 5,000 FLIGHTS

Operational reliability diverges sharply across airlines, while passenger utilization and overall satisfaction ratings remain clustered.

CARRIER	FLIGHTS	AVG DELAY	CANCELLATION	REVENUE	LOAD FACTOR
IndiGo	1,780	14.7 min	0.84%	₹1.55 B	82%
Air India	1,120	17.2 min	1.45%	₹1.02 B	81%
SpiceJet	890	21.0 min	3.18%	₹0.68 B	82%
Vistara	750	15.4 min	1.02%	₹0.64 B	82%
Akasa Air	460	16.1 min	1.20%	₹0.42 B	81%

FLEET LEADER

IndiGo volume & turn discipline

Captured 35.6% of flight volume with sub-15 min arrival delay and lowest cancellation (0.84%).

RISK OUTLIER

SpiceJet turnaround exposure

Highest delay (21.0 min) and cancellation rate (3.18%) indicate station turn vulnerability.

RATING INVARIANCE

Customer rating: ~3.7 / 5

Correlation between delay and customer score was r = 0.01, showing ratings reflect broader factors.

Made with GAMMA

ACTIONABLE NEXT STEPS

RECOMMENDED OPERATIONAL PRIORITIES

Proposed scheduling, weather and route actions should be validated before rollout.

Evaluate evening bank buffers (7:00–9:00 PM)

Test 10–15 minutes of additional turnaround buffer or dedicated ground handlers.

Prepare weather protocols

Plan slot swaps and rebooking for storms and fog. Storm delays averaged 39 minutes.

Investigate performance and routes

Investigate SpiceJet performance and audit Air India's Hyderabad–Delhi and SpiceJet's Mumbai–Bengaluru routes.

Validate before rollout

Treat these as proposals, not confirmed causes or measured improvements; test before changing published block times.



Made with GAMMA